

Rescue3D — How It Works

A 3D Interactive Disaster Management and Emergency Response Simulator
Nilusha Madhuwanthi Jayasekara | September 2026

1 The Real-World Problem

Emergency management training faces three persistent gaps that Rescue3D addresses:

- **Consequence-free learning is missing.** Real emergencies cannot be paused or replayed. Trainees need a sandbox where mistakes teach rather than cost lives.
- **Dashboards exist, decisions do not.** Most simulation tools show data but never ask the operator to act on it. Rescue3D makes the operator dispatch units, assign routes and watch outcomes unfold.
- **No safe place to be wrong.** An operator who has never handled a chemical spill at a warehouse next to a residential block will freeze the first time. Rescue3D lets them rehearse that exact scenario in a 3D city.

2 Key Design Decisions

Decision	What	Why
Interactive 3D city engine	Operators see the city from any angle, zoom into buildings, and watch disaster effects spread in real time. Built with React Three Fiber / Three.js.	Spatial awareness and immersion make training stick longer than flat dashboards.
Role-based JWT authentication	Five roles (administrator, coordinator, operator, analyst, viewer) with permission-based route guards and secure token rotation.	Each trainee sees only the controls their real-world role would have.
Deterministic routing engine	Haversine distance, severity-driven risk scoring in [0,1], travel-time estimation inversely proportional to road risk. Pure functions in packages/simulation-core.	Reproducible scenarios: same inputs always produce the same route, so instructors can compare trainee decisions.
Real-time Socket.IO sync	Incident creation, status changes and unit dispatches broadcast instantly to all connected operators via Socket.IO.	Multi-operator drills work without manual page reloads.
Shared TypeScript contracts	packages/contracts defines every API payload, incident type, severity level and scene route interface. Both frontend and backend import the same types.	Compile-time guarantees that the 3D scene, REST API and UI always agree on shapes.
Monorepo with clean ownership	apps/web, apps/api, packages/contracts, packages/simulation-core, packages/ui, packages/config. Each package builds independently.	Two developers work in parallel without merge conflicts. Nilusha owns API, incidents, routing, analytics, auth; Kavindu owns the 3D scene and deployment.
Accessibility-first controls	System / Dark / Light theme switcher, Low-Motion and Reduced-Quality toggles for the 3D viewport.	Inclusive for users with vestibular or visual sensitivities.

3 Architecture

Rescue3D is a TypeScript monorepo with two deployable applications and four shared packages:

Layer	Technology	Deployed to
Frontend (apps/web)	React 19, TypeScript, Vite, React Three Fiber, Tailwind CSS, TanStack Query, Zustand, React Hook Form, Zod, Recharts, Framer Motion, Socket.IO Client	GitHub Pages
Backend (apps/api)	Node.js, Express, TypeScript, MongoDB/Mongoose, Socket.IO, JWT (access + refresh), bcrypt, Zod validation, Swagger/OpenAPI, Helmet	Render (free tier)
Shared contracts	TypeScript interfaces — incident types, severity, unit status, scene route, API payloads	npm workspace (build-time)
Simulation core	Haversine distance, risk scorer, route planner, recommendation engine	npm workspace (build-time)

4 Nilusha's Scope

Role: Full-Stack Product, Incident Intelligence & Emergency Response Lead

- **Authentication & operator access** — JWT access + refresh token rotation, five roles with permission guards, bcrypt hashing, Zod validation, inline field errors.
- **Incident Command Center** — report incidents (fire, flood, earthquake, medical, hazmat, structural), status lifecycle (reported - dispatched - in_progress - contained - resolved), real-time unit assignment, operator cards with distance/ETA/risk.
- **Deterministic routing engine** — Haversine distance, severity-driven risk scoring, travel-time estimation, waypoint interpolation producing {nodeId, x, y, z} for the 3D scene.
- **Analytics dashboard** — average response time, containment rate, resource utilization, incident breakdown by status and severity, unit status tracking.
- **Real-time synchronization** — Socket.IO hooks (useRealtimeIncidents) for instant cross-operator updates.
- **Quality & testing** — 79-test automated suite (52 API + 27 Web), TypeScript strict-mode type checking, code review process, QA report.
- **Design system** — color tokens (canvas, surface, ember, jade, danger, warning, success), button variants (Primary, Secondary, Danger, Ghost, Disabled), badges, consistent theming.

5 Application Screenshots

Captured from the live deployment at nilushamadhuwanthi123.github.io/rescue3d-disaster-response-simulator connected to the Render-hosted API.

Create an account

Already have one? [Sign in](#)

Name

Name is required

Email

Password

- At least 8 characters
- An uppercase letter
- A number

Create account

Fig 1 — Create an account. Password rules (8+ chars, uppercase, number) are stated up front and validate in real time. The same rules are enforced server-side.

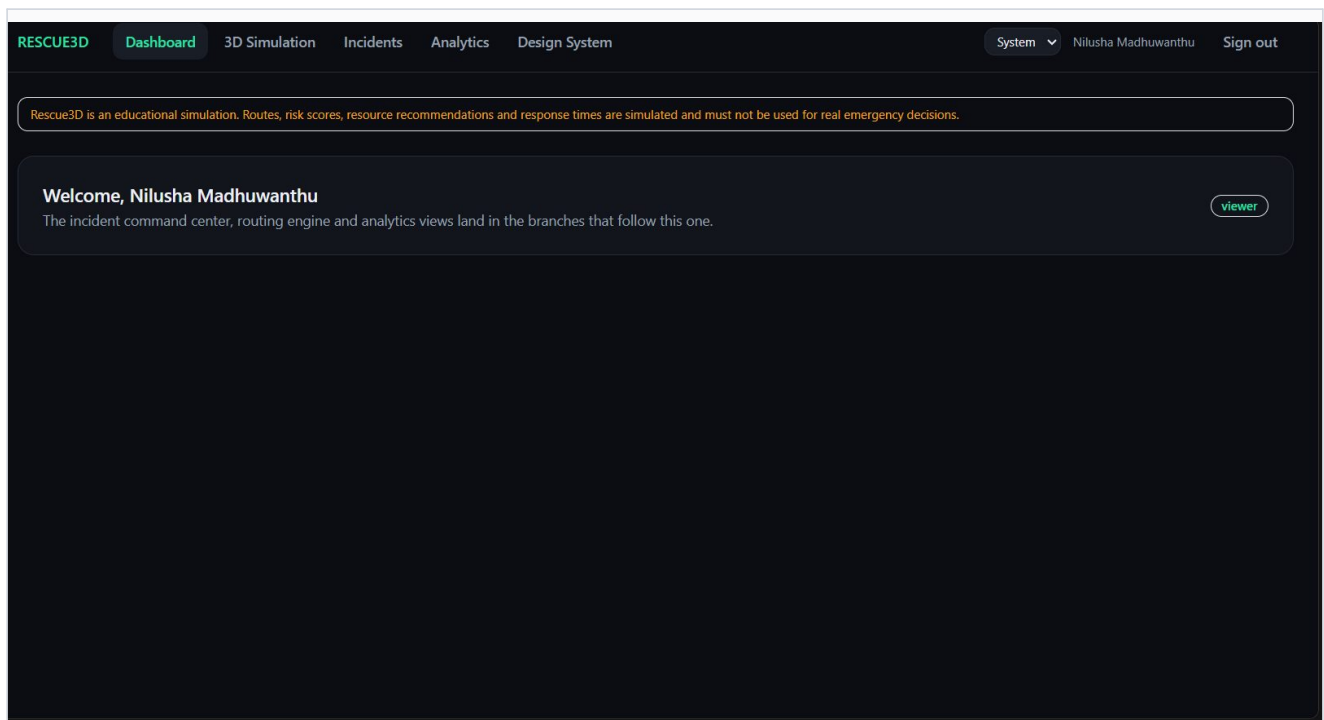


Fig 2 — Dashboard. Welcome screen with operator name, role badge (viewer), educational simulation disclaimer, and navigation to all modules.

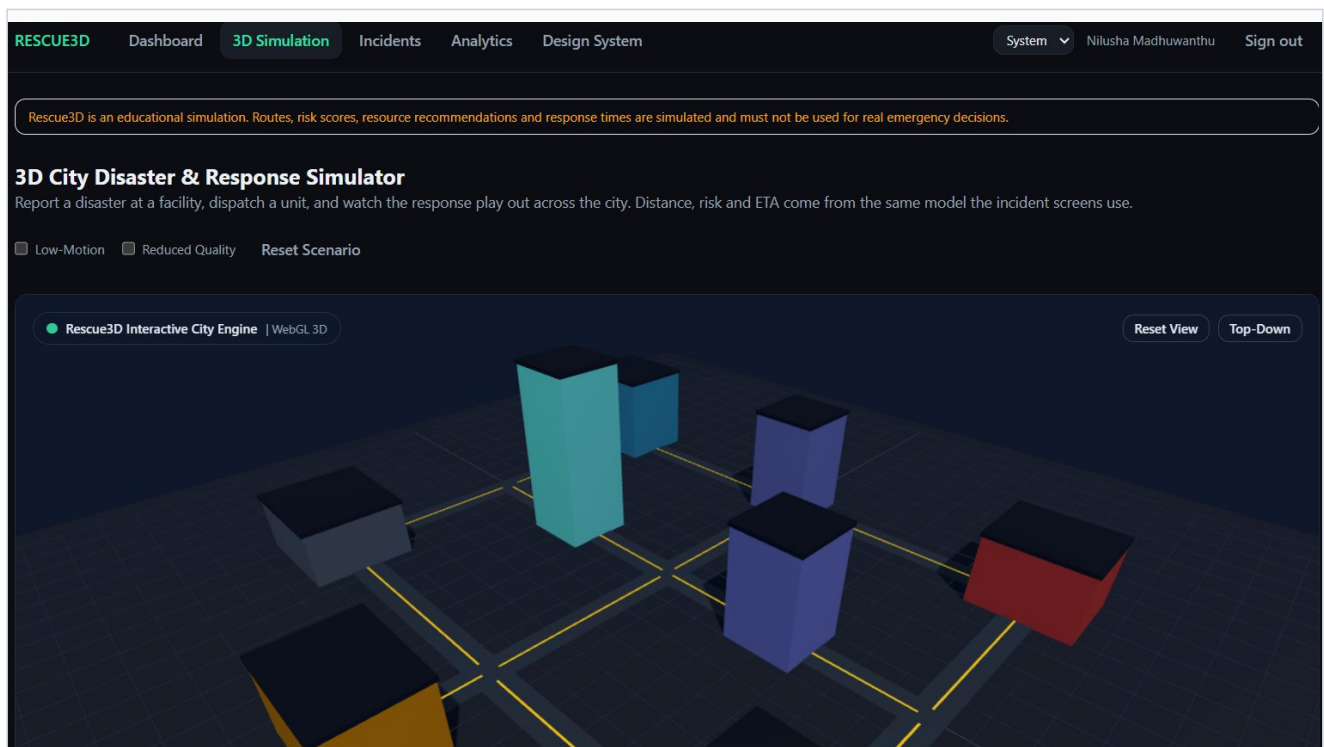


Fig 3 — 3D Simulation. The interactive WebGL city engine renders buildings, roads and grid lines. Reset View and Top-Down controls adjust the camera.

1. Select Target Building

Metro Financial Tower
Commercial

Harbor Logistics Warehouse
Industrial

Silicon Bay Tech Campus
Commercial

City Civic Auditorium
Commercial

Grand Plaza Hotel & Suites
Residential

2. Configure Disaster

Disaster Type

Fire

Flood

Earthquake

Severity Level

Low

Moderate

High

Critical

Trigger FIRE Incident

3. Emergency Dispatch

Trigger a disaster above to activate emergency dispatch recommendations and waypoint route navigation.

Fig 4 — 3D Simulation controls. Select a target building (Metro Financial Tower, City Civic Auditorium, Harbor Logistics Warehouse, Grand Plaza Hotel, Silicon Bay Tech Campus), choose disaster type (Fire / Flood / Earthquake) and severity (Low / Moderate / High / Critical), then trigger the incident.

RESCUE3D

Dashboard

3D Simulation

Incidents

Analytics

Design System

System

Nilusha Madhuwanthu

Sign out

Rescue3D is an educational simulation. Routes, risk scores, resource recommendations and response times are simulated and must not be used for real emergency decisions.

Active incidents

No incidents reported yet.

Report an incident

Title

Description

Type

fire

fire

flood

earthquake

medical

hazmat

structural

Severity

moderate

Longitude

79.8612

Report incident

Fig 5 — Incidents page with incident type dropdown. Six incident types: fire, flood, earthquake, medical, hazmat, structural. The form includes title, description, coordinates and a Report button.

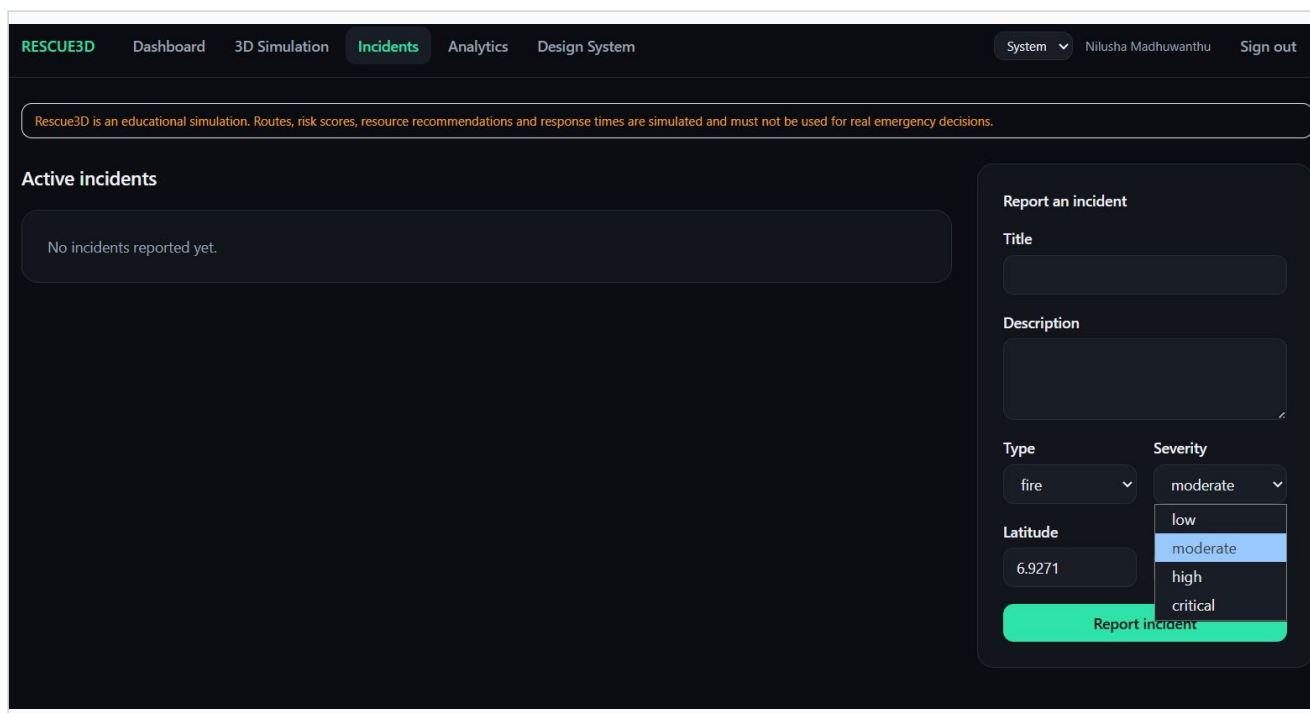


Fig 6 — Incidents page with severity dropdown. Four severity levels: low, moderate, high, critical.

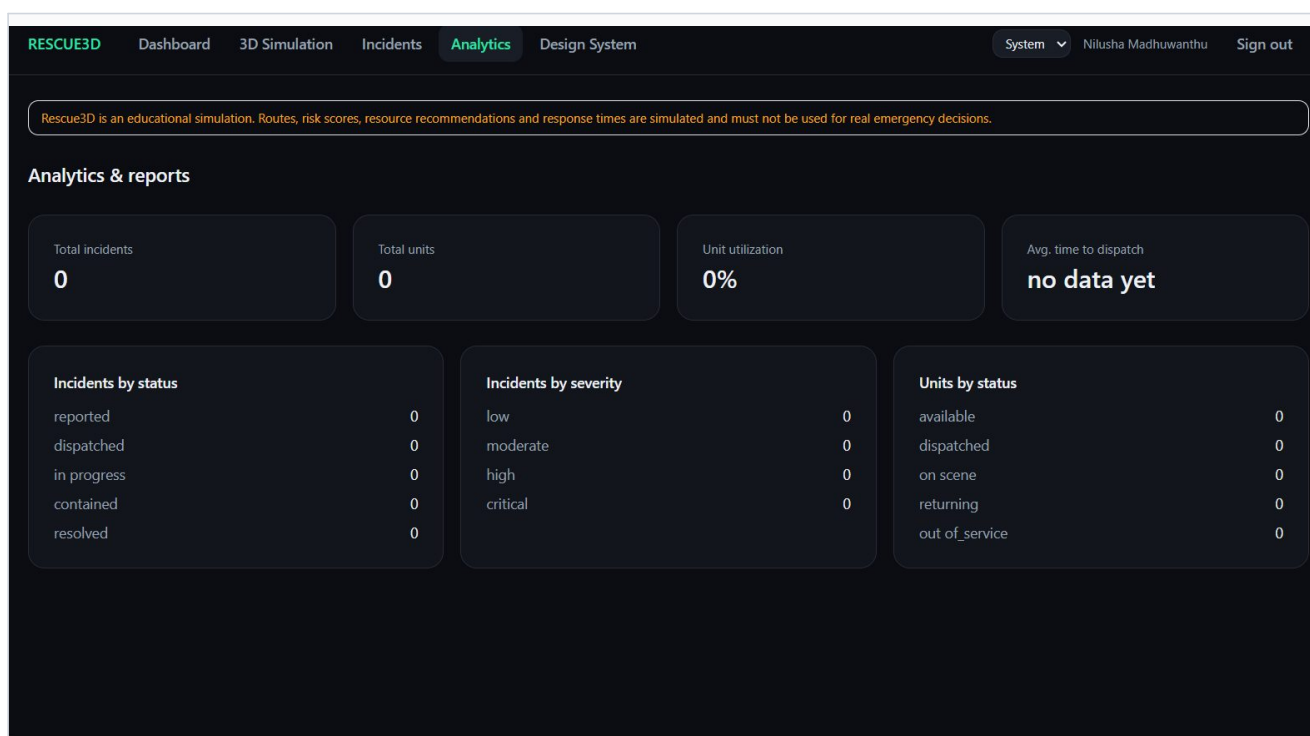


Fig 7 — Analytics & reports. KPI cards (Total incidents, Total units, Unit utilization, Avg. time to dispatch), breakdowns by incident status, severity and unit status.

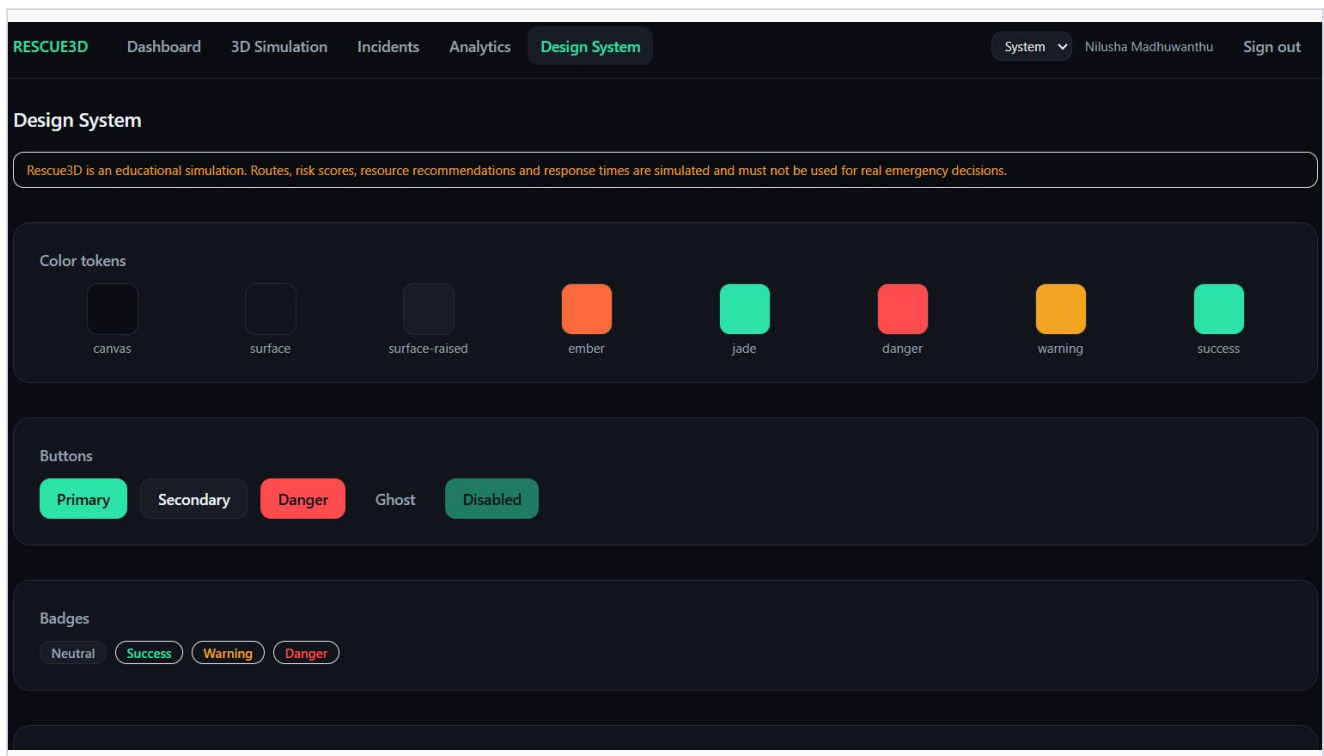


Fig 8 — Design System. Color tokens (canvas, surface, surface-raised, ember, jade, danger, warning, success), button variants and badge styles. The system ensures visual consistency.

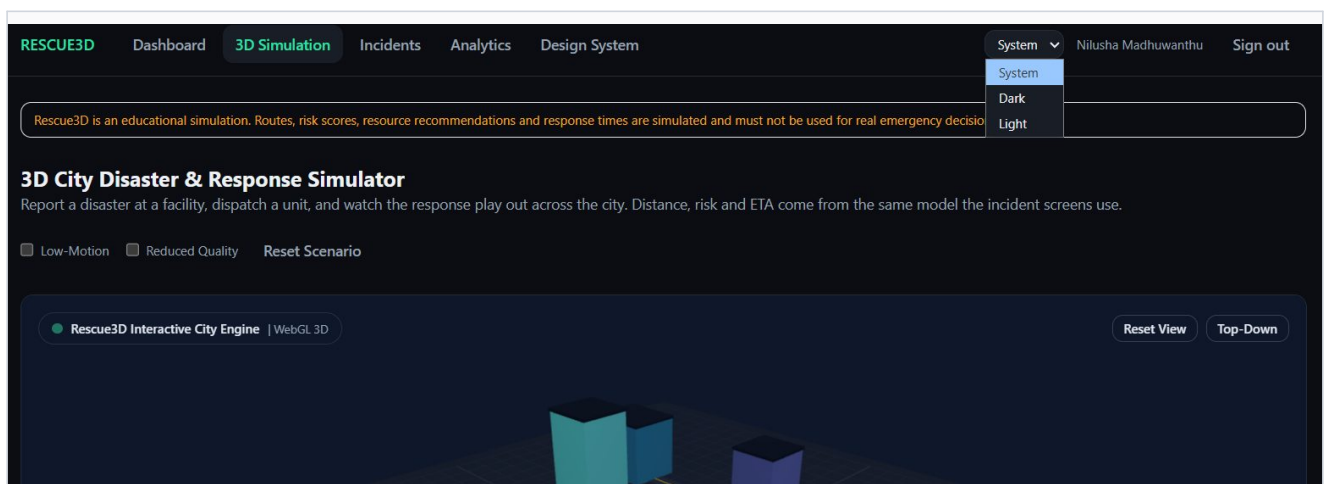


Fig 9 — Theme switcher. System / Dark / Light modes for accessibility. The Low-Motion and Reduced-Quality toggles are also visible for the 3D viewport.

6 Quality Position

Metric	Result
Automated tests	79 / 79 pass (52 API + 27 Web)
TypeScript strict-mode	Clean (zero errors)
Production build	Success (Vite)
Security headers	Helmet defaults active

Metric	Result
Defects found & fixed	7 fixed in review (D-02 to D-08)
Open defects	1 (D-01: ESLint config missing, CI unaffected)
Code review	All PRs (#1-#7) reviewed before merge

7 Roadmap

- **End-to-end Playwright tests** — login, incident reporting and dispatch flows.
- **ESLint configuration** — resolve D-01 by adding eslint.config.ts and the ESLint dependency.
- **Populated analytics demo** — seed the database with sample incidents so the analytics dashboard shows realistic KPIs in the demo.
- **Route visualization on 3D city** — draw the calculated safe route as an animated path through the city streets.
- **Multi-operator drill mode** — two or more operators join the same scenario and see each other's dispatches in real time.
- **Mobile-responsive incident cards** — field operators on tablets or phones.